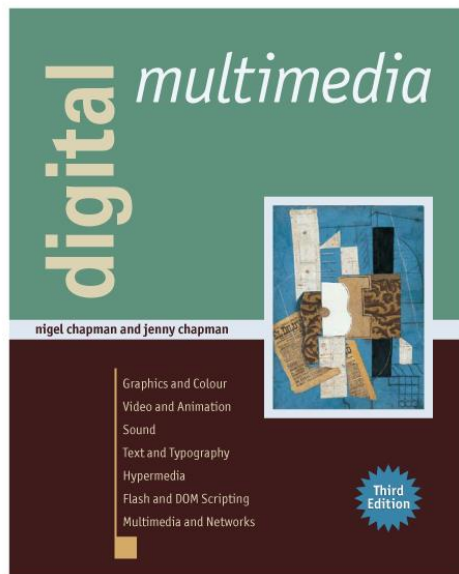




# 4

## Bitmapped Images

**Based on material from  
*Digital Multimedia*, 3rd edition  
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# Resolution

**Resolution is a measure of how finely a device approximates continuous images using finite pixels.**

**The resolution of scanners and printers is usually equated with their pixel density (dots per inch).**

**The resolution of video frames and computer monitors is usually equated with their pixel dimensions (width  $\times$  height).**

**The resolution of a digital still camera is often quoted as the total number of pixels in the largest image it can record.**

**The pixel density of monitors and still cameras is assumed to be 72 dpi ("screen resolution").**

## 4

## BITMAPPED IMAGES

## RESOLUTION 136

**A bitmapped image has pixel dimensions but no intrinsic pixel density.**

## 4

## BITMAPPED IMAGES

## RESOLUTION 137

**The physical size of an image when it is displayed will depend on the pixel density of the device it is to be displayed on.**

## 4

## BITMAPMED IMAGES

## RESOLUTION 138

72 dpi



115 dpi



600 dpi

*Device resolution and image size*

## 4

## BITMAPMED IMAGES

## RESOLUTION 139

**Most image formats record a resolution (pixel density) together with the image data; this is usually the resolution of the device on which it originated.**

## 4

## BITMAPMED IMAGES

RESOLUTION 140

72 dpi, 198 × 149 px



600 dpi, 1654 × 1240 px

*Resolution and pixel dimensions*

## 4

## BITMAPMED IMAGES

RESOLUTION 141

**Scaling an image without changing its resolution changes its pixel dimensions in the same way as changing its resolution without changing its size.**

## 4

## BITMAPPED IMAGES

## RESOLUTION 142

**Reducing the pixel dimensions is called “downsampling”; increasing them is called “upsampling”. Both can lead to a loss of quality.**

## 4

## BITMAPPED IMAGES

## RESOLUTION 143

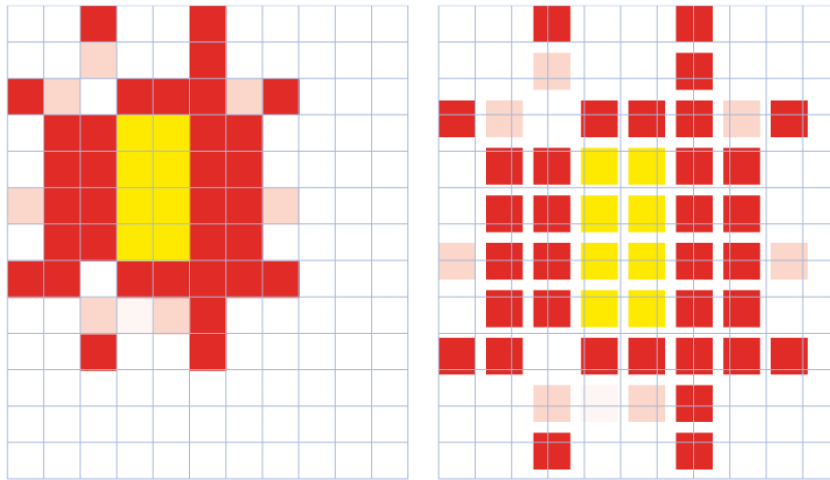
**Scaling can be done either by applying a transformation to each original pixel or by applying the inverse transformation to each pixel in the scaled image.**

**Interpolation is needed because of the finite size of pixels.**

## 4

## BITMAPMED IMAGES

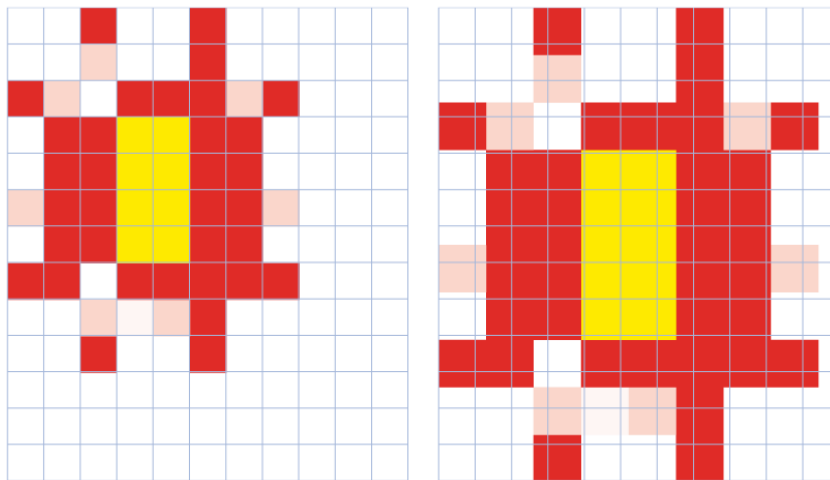
RESOLUTION 144

*Scaling pixels*

## 4

## BITMAPMED IMAGES

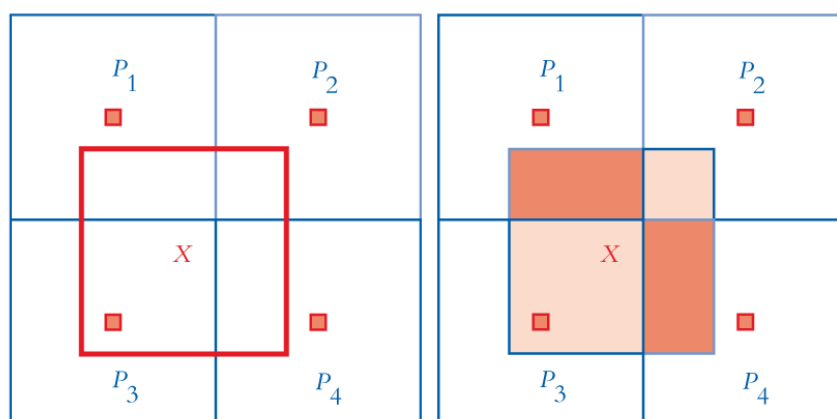
RESOLUTION 145

*Scaling pixels using a reverse mapping*



**Nearest-neighbour, bilinear or bicubic interpolation may be used.**

**Nearest-neighbour interpolation is quickest but produces poor-quality results; bicubic is slowest but produces very good results; bilinear is in between.**



*Pixel interpolation*

*Areas for bilinear interpolation*

## 4

## BITMAPPED IMAGES

RESOLUTION 148



*Nearest-neighbour (left), bilinear (middle) and bicubic (right) interpolation*

## 4

## BITMAPPED IMAGES

RESOLUTION 149

**Interpolation causes a loss of information which can never be recovered.**

# Image Compression

**Images can be losslessly compressed using various methods, including run-length encoding (RLE), Huffman encoding and the dictionary-based LZ77, LZ78, LZW and deflate algorithms.**

**JPEG is the most commonly used lossy compression method for still images.**

**High-frequency information can be discarded from an image without perceptible loss of quality, because people do not perceive the effects of high frequencies in images very accurately.**

**The image is mapped into the frequency domain using the Discrete Cosine Transform (DCT).**

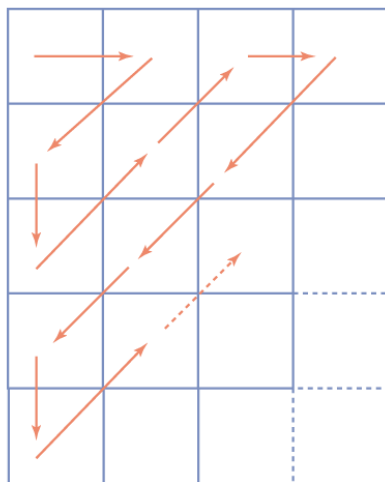
**The Discrete Cosine Transform is applied to  $8 \times 8$  blocks of pixels.**

**Applying the DCT does not reduce the size of the data, since the array of frequency coefficients is the same size as the original pixel array.**

**The coefficients are quantized, according to a quantization matrix which determines the quality. The quantization discards some information.**

**After quantization there will usually be many zero coefficients. These are RLE-encoded, using a zig-zag sequence to maximize the length of the runs.**

**The non-zero coefficients are compressed using Huffman encoding.**



*The zig-zag sequence*

**Decompression is performed by reversing the process, using the Inverse DCT to recover the image from its frequency domain representation.**

**The decompressed image may exhibit compression artefacts, including blurring and visible edges at the boundaries between the  $8 \times 8$  pixel blocks, especially at low quality settings.**

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## BITMAPPED IMAGES

## IMAGE COMPRESSION 160



*Original (left) and JPEG (right)*

## 4

## BITMAPPED IMAGES

## IMAGE COMPRESSION 161

**JPEG2000 improves on JPEG in many areas, including image quality at high compression ratios. It can be used losslessly as well as lossily.**



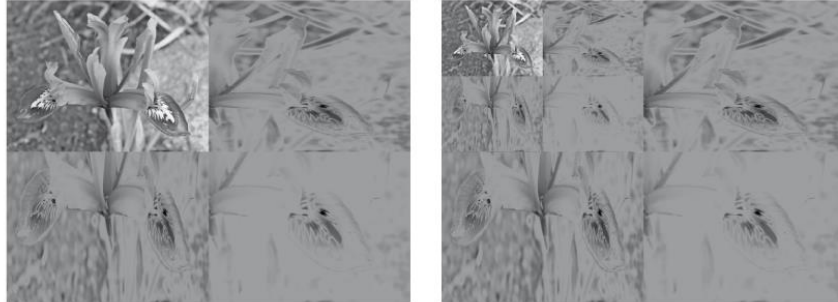
**For JPEG2000 compression the image is divided into tiles, but these can be any size, up to the entire image.**

**A Discrete Wavelet Transform (DWT) is applied to the tiles, generating a wavelet decomposition, comprising a coarse (low resolution) version of the image and a set of detail coefficients that can be used to add progressively more detail to the image.**

## 4

BITMAPPED IMAGES

IMAGE COMPRESSION 164



*Wavelet decomposition of an image*

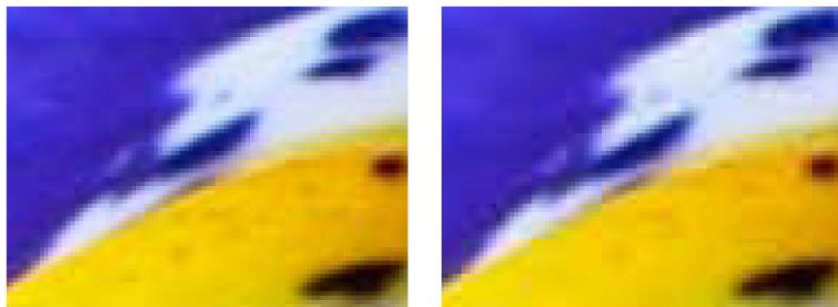
## 4

BITMAPPED IMAGES

IMAGE COMPRESSION 165

**The DWT may be reversible (lossless) or irreversible (lossy).**

**The detail coefficients in the wavelet decomposition may be quantized and are then losslessly compressed using arithmetic encoding.**



*Original (left) and JPEG2000 (right)*

## File Formats

**GIF files use LZW compression and are restricted to 256 colours. One colour may be used to designate transparency.**

**They are most suitable for simple images with areas of flat colour.**

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## BITMAPPED IMAGES

## FILE FORMATS 170



*A GIF with transparent areas*

## 4

## BITMAPPED IMAGES

## FILE FORMATS 171

**PNG was developed to supersede GIF. It uses deflate compression, is not restricted to 256 colours and supports alpha channels for partial transparency.**

**JPEG data can be stored in several different formats.**

**JFIF and SPIFF are compatible formats for JPEG images and are widely used on the Web.**

**Exif can hold either JPEG or TIFF data, together with extensive metadata.**

**JP2 and JPX formats are defined for storing JPEG2000 data.**

**TIFF is an extensible format, often used for storing uncompressed digital photographs, and for interchange of images.**

**BMP is a simple bitmapped image format that is native to Windows, but widely supported. BMP files are often uncompressed.**

**PDF documents can include bitmapped image data, that may be compressed using JPEG, JPEG2000, LZW, deflate, and others.**

**Camera raw data is used when complete control over image processing is required, but there is no standard format for camera raw data.**

**Adobe's DNG (Digital Negative) format is a standard, based on the TIFF format, intended for archiving camera raw images.**



**Image file formats differ in their support for metadata.**